



**DEPARTAMENTO DE ELETRÓNICA, TELECOMUNICAÇÕES
E INFORMÁTICA**

MESTRADO EM ENGENHARIA DE COMPUTADORES E TELEMÁTICA

ANO 2026/2027

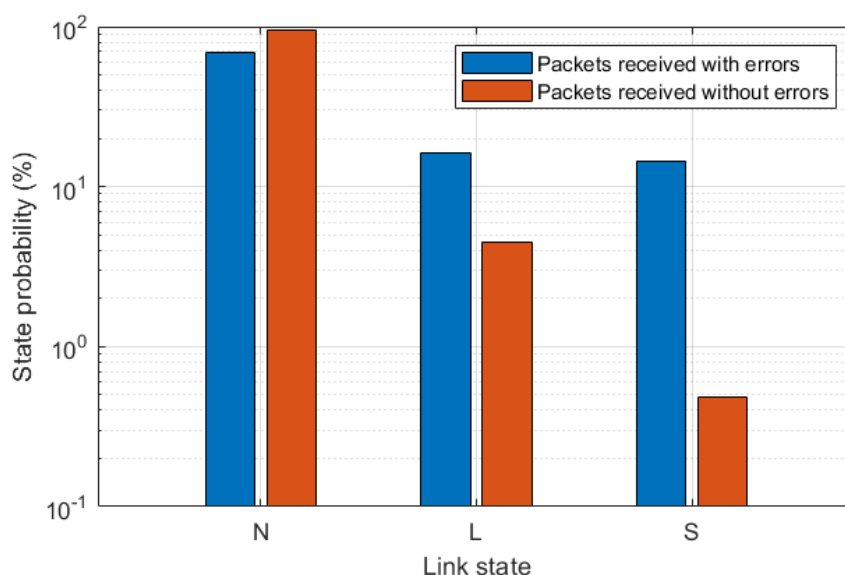
MODELAÇÃO E DESEMPENHO DE REDES E SERVIÇOS

PRACTICAL GUIDE NO. 1

Task 1

Consider a wireless link for data communications that can be in one of three possible states: Normal (N) with probability 95%, Light Interference (L) with probability 4.5% or Severe Interference (S) with probability 0.5%. The probability of the transmitted data packets being received with errors is 0.1% (when the link is in state N), 0.5% (when the link is in state L) and $p_S = 4\%$ (when the link is in state S). The receiver hosts detect with 100% probability if data packets are or are not received with errors.

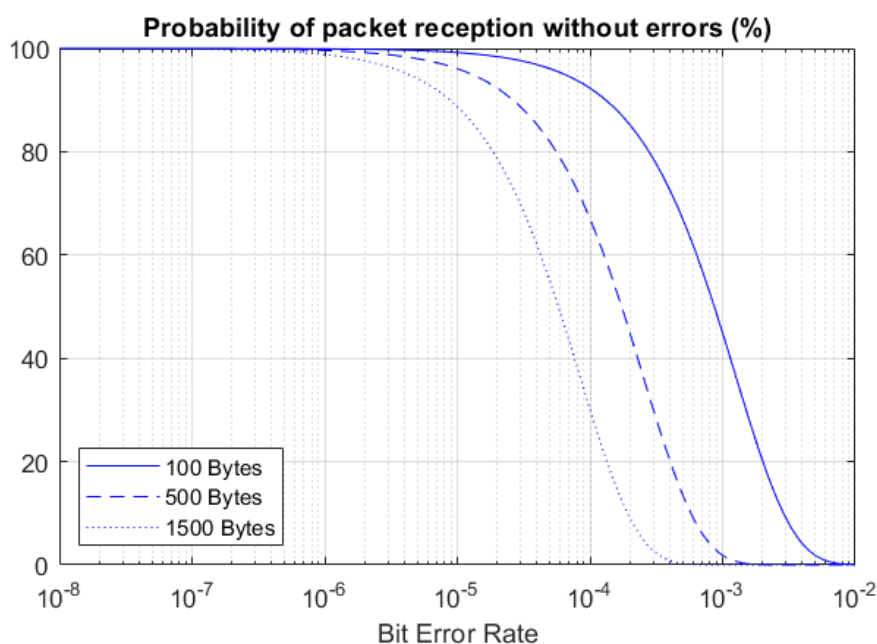
- 1.a. Determine the probability of a packet being received with errors and the link being in state N . Answer: 0.095%
- 1.b. Determine the probability of packets being received with errors. Answer: 0.137%
- 1.c. Determine the probability of the link being in state S when a packet is received with errors. Answer: 14.545%
- 1.d. Determine the probability of the link being in state L when a packet is received without errors. Answer: 4.484%
- 1.e. Draw a plot (with the same look as the plot below) with the probability of each link state when the data packets are received with errors and without errors. What do you conclude from these results?



Task 2

Consider a wireless link for data communications with a bit error rate (*ber*) of p . Assume that transmission errors in the different bits of a data frame are statistically independent (i.e., the number of errors of a data packet is a binomial random variable).

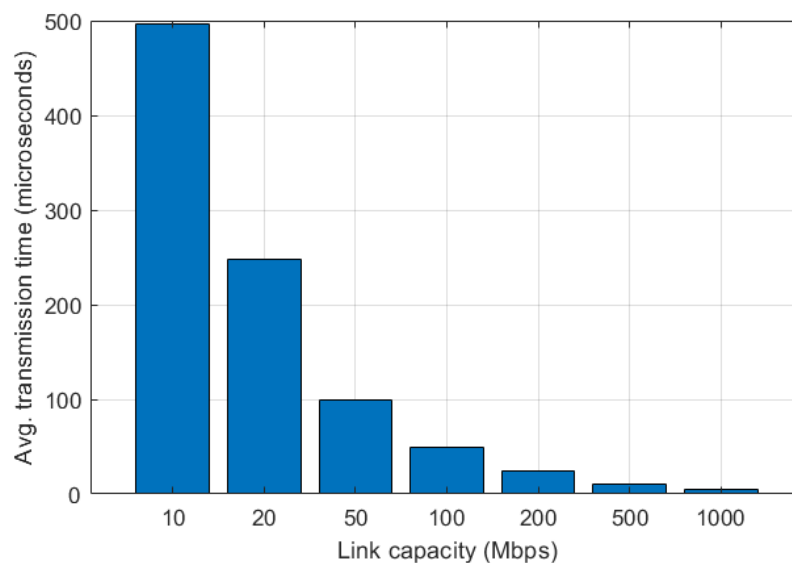
- 2.a. Determine the probability of a data frame of 500 Bytes to be received without errors when $p = 10^{-3}$. Answer: 1.828%
- 2.b. Determine the probability of a data frame of 1000 Bytes to be received with exactly one error when $p = 10^{-4}$. Answer: 35.949%
- 2.c. Determine the probability of a data frame of 1500 Bytes to be received with one or more errors when $p = 10^{-5}$. Answer: 11.308%
- 2.d. When $p = 10^{-5}$, what is the maximum data frame size so that the probability of the frames being received with one or more errors is not higher than 1%? Answer: 125 Bytes
- 2.e. Draw a plot using a logarithmic scale for the X-axis (use the MATLAB function `semilogx`) with the same look as the plot below with the probability of a data frame (of size 100 Bytes, 500 Bytes or 1500 Bytes) being received without errors as a function of the *ber* (from $p = 10^{-8}$ up to $p = 10^{-2}$). What do you conclude from these results?



Task 3

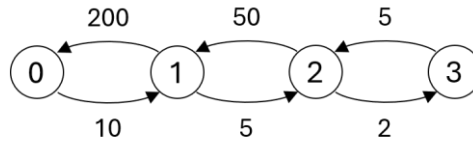
Consider a data link of capacity C supporting a flow of data packets, whose sizes are between 64 and 1518 bytes with the probabilities: 19% for 64 bytes, 23% for 110 bytes, 17% for 1518 bytes and an equal probability for all other values (i.e., from 65 to 109 and from 111 to 1517). Consider the random variable S representing the packet transmission time.

- 3.a.** Determine the average, the second moment, the variance and the standard deviation of the random variable S when the link capacity is $C = 10$ Mbps. Answer: $E[S] = 0.496$ msec, $E[S^2] = 0.463$ msec², $Var[S] = 0.217$ msec², $\sigma[S] = 0.466$ msec
- 3.b.** Determine the probability of S being higher than 1 millisecond when the link capacity is $C = 10$ Mbps. Answer: 24.54%
- 3.c.** Repeat **3.a** but now considering that the link capacity is $C = 100$ Mbps. What do you conclude? Answer: $E[S] = 0.0496$ msec, $E[S^2] = 0.00463$ msec², $Var[S] = 0.00217$ msec², $\sigma[S] = 0.0466$ msec
- 3.d.** Determine the probability of S being higher than 1 millisecond when the link capacity is $C = 100$ Mbps. What do you conclude? Answer: 0.00%
- 3.e.** Draw a plot (with the same look as the plot below) with the average packet transmission time when the link capacity is $C = 10, 20, 50, 100, 200, 500$ and 1000 Mbps. What do you conclude from these results?



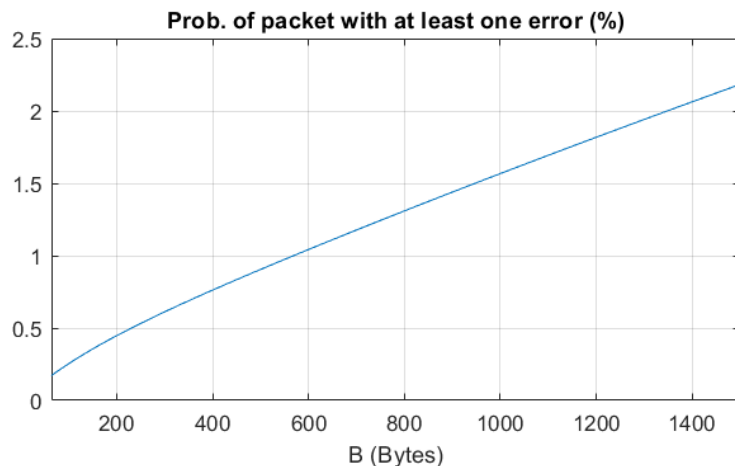
Task 4

Consider a wireless link between multiple stations for data communications with states $i = 0, 1, 2$ and 3 whose behavior is given by the following CTMC (Continuous Time Markov Chain), such that transition rates are in transitions per hour:



The bit error rate (*ber*) introduced by the wireless link (due to the variation of the propagation and interference factors along with time) on each state is $10^{-6} \times 10^i$. All stations detect with a probability of 100% when the data frames sent by the other stations are received with errors.

- 4.a. Determine the probability (in %) of the link being in each state. Answer:
 94.6% ($i = 0$), 4.73% ($i = 1$), 0.473% ($i = 2$), 0.189% ($i = 3$)
- 4.b. Determine the average percentage of time the link is in each state. Answer:
 94.6% ($i = 0$), 4.73% ($i = 1$), 0.473% ($i = 2$), 0.189% ($i = 3$)
- 4.c. Determine the average *ber* of the link. Answer: 3.78×10^{-6}
- 4.d. Determine the average holding time (in minutes) of the link in each state. Answer:
 6.0 min ($i = 0$), 0.293 min ($i = 1$), 1.154 min ($i = 2$), 12.0 min ($i = 3$)
- 4.e. Determine the probability (in %) of a data frame of 1500 Bytes being received with no errors in each state. Answer:
 98.81% ($i = 0$), 88.69% ($i = 1$), 30.12% ($i = 2$), 0.001% ($i = 3$)
- 4.f. Determine the probability of a data frame of 1500 Bytes being received with no errors. Answer: 97.82%
- 4.g. Determine the probability of the link being in each state when a data frame of 1500 Bytes is received with no errors. Answer:
 95.57% ($i = 0$), 4.29% ($i = 1$), 0.146% ($i = 2$), 0.000% ($i = 3$)
- 4.h. Determine the probability of the link being in each state when a data frame of 1500 Bytes is received with at least one error. Compare these results of the ones in 4.a and 4.g and draw conclusions. Answer:
 51.69% ($i = 0$), 24.50% ($i = 1$), 15.142% ($i = 2$), 8.667% ($i = 3$)
- 4.i. Draw a plot with the same look as the plot on the right of the probability of the packet being received with at least one error as a function of the packet size (from 64 Bytes up to 1500 Bytes). Analyze and justify the results.



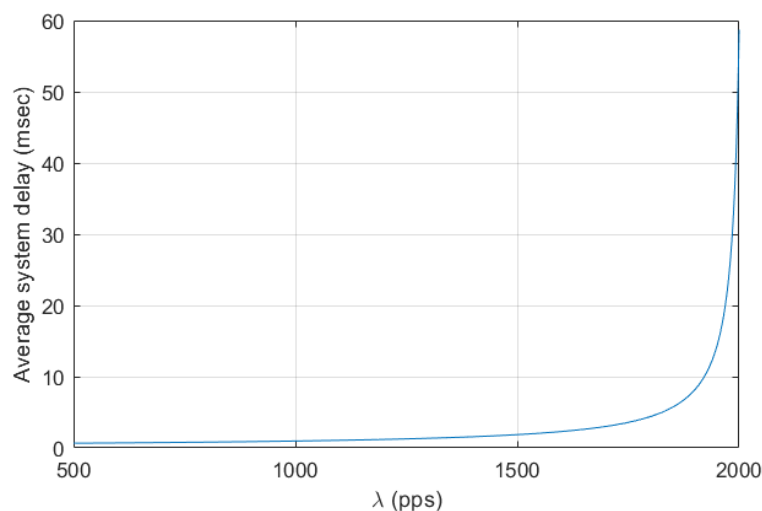
Task 5

Consider an ideal point-to-point link (i.e., with a $ber = 0$) from one router to another router with a capacity of C Mbps (1 Mbps = 10^6 bps) for IP communications. There is a very large queue at the output port of the link. The link has a propagation delay of $10\ \mu\text{s}$ ($1\ \mu\text{s} = 10^{-6}$ seconds).

The link supports a flow of IP packets, whose sizes are between 64 and 1518 bytes with the probabilities: 19% for 64 bytes, 23% for 110 bytes, 17% for 1518 bytes and an equal probability for all other values (i.e., from 65 to 109 and from 111 to 1517). The IP packet arrivals is a Poisson process with rate λ pps (packets/second).

Consider that $\lambda = 1500$ pps and $C = 10$ Mbps.

- 5.a. Determine the average packet size (in Bytes) and the average packet transmission time (in milliseconds) of the IP flow. Answer: 620.02 Bytes, 0.496 msec.
- 5.b. Determine the throughput (in Mbps) of the link. Answer: 7.44 Mbps
- 5.c. Determine the capacity of the link, in packets/second. Answer: 2016.06 pps
- 5.d. Determine the average packet queuing delay and the average packet system delay of the IP flow (the system delay is the queuing delay + transmission time + propagation delay) using **the M/G/1 queuing model**. Answer: queuing – 1.358 msec, system – 1.864 msec
- 5.e. Determine the average packet queuing delay and the average packet system delay of the IP flow using now **the M/M/1 queuing model**. Compare these results with the previous ones in 5.d and draw conclusions. Answer: queuing – 1.442 msec, system – 1.948 msec
- 5.f. Using **the M/M/1 queuing model**, determine the probability of the queue occupation being larger than P packets, for $P = 10, 20$ and 30 . What do you conclude? Answer: 2.878% ($P=10$), 0.150% ($P=20$), 0.008% ($P=30$)
- 5.g. Draw a plot with the same look as the plot below with the average packet system delay given by **the M/G/1 queuing model** as a function of the packet arrival rate λ (from $\lambda = 500$ pps up to $\lambda = 2000$ pps). Analyze the results and draw conclusions.



Task 6

Consider a point-to-point link with a $ber = 10^{-5}$ from one router to another router with a capacity of C Mbps for IP communications. There is a very large queue at the output port of the link. The link has a propagation delay of $10 \mu s$.

The link supports a flow of IP packets, whose sizes are between 64 and 1518 bytes with the probabilities: 19% for 64 bytes, 23% for 110 bytes, 17% for 1518 bytes and an equal probability for all other values (i.e., from 65 to 109 and from 111 to 1517). The IP packet arrivals is a Poisson process with rate λ pps (packets/second).

Consider that $\lambda = 18000$ pps and $C = 100$ Mbps.

- 6.a.** Determine the average packet size (in Bytes) of the IP flow. Answer: 620.02 Bytes
- 6.b.** Determine the packet loss rate (in %) of the link. Answer: 4.7365%
- 6.c.** Determine the packet loss rate (in %) of the link considering only the average packet size determined in **6.a**. Compare this result with the previous one in **6.b** and draw conclusions. Answer: 4.8392%
- 6.d.** Determine the throughput and the goodput of the link (both in Mbps). Answer: 89.28 Mbps, 81.37 Mbps
- 6.e.** Determine the average packet system delay of the IP flow (the system delay is the queuing delay + transmission time + propagation delay). Answer: 0.449 msec

Task 7

Consider an ideal point-to-point link from one router to another router with a capacity of C Gbps ($1 \text{ Gbps} = 10^9 \text{ bps}$) for IP communications. There is a very large queue at the output port of the link. The link has a propagation delay of $1 \mu\text{s}$.

The link supports 2 flows of IP packets. The packet sizes of IP flow 1 are between 64 and 200 bytes (the same probability for all values) and the packet sizes of IP flow 2 are between 100 and 1500 bytes (the same probability for all values). The IP packet arrivals is a Poisson process with rate λ_1 pps for IP flow 1 and λ_2 pps for IP flow 2.

Consider that $\lambda_1 = 4 \times 10^4$ pps, $\lambda_2 = 1.4 \times 10^5$ pps and $C = 1 \text{ Gbps}$.

- 7.a. Consider the random variable S representing the transmission time of the aggregate of the two IP flows. Determine the average $E[S]$ and the second moment $E[S^2]$. Answer: $E[S] = 5.212 \times 10^{-6} \text{ sec}$, $E[S^2] = 4.027 \times 10^{-11} \text{ sec}^2$
- 7.b. Determine the average packet system delay of the IP flow 1 (the system delay is the queuing delay + transmission time + propagation delay) when both flows are statistically multiplexed in the link. Answer: $60.74 \mu\text{sec}$
- 7.c. Determine the average packet system delay of the IP flow 1 when 50 Mbps of the link are assigned to IP flow 1. Answer: $117.02 \mu\text{sec}$
- 7.d. Determine the average packet system delay of the IP flow 1 when 100 Mbps of the link are assigned to IP flow 1. Compare this result with the previous ones in 7.b and 7.c and draw conclusions. Answer: $9.779 \mu\text{sec}$
- 7.e. Draw a plot with the same look as the plot below with the average packet system delay of IP flow 1 when this flow is assigned with 50, 75, 100, 125 and 150 Mbps. Analyze the results and draw conclusions.

